OPTICS · SEMESTER 5 · MODULE 4

Answer key: Polarisation

Formal solutions to all 42 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey mark notes show a typical split of marks (a guide, not an official scheme).

Section A · Short answers

2 marks each.

1 PYQ N24 Q10

Define polarization.

KNOW FIRST

T1 Polarisation and its kinds P1 Waves: the words you need P2 Light is an electromagnetic wave

Polarisation is the phenomenon by which the vibrations of a transverse wave (for light, the electric field vector) are restricted to a particular direction, or a definite pattern, in the plane perpendicular to the direction of propagation. Light in which the electric vector vibrates along a single fixed direction is called plane (linearly) polarised.

2

Plane of vibration vs plane of polarisation.

KNOW FIRST

T1 Polarisation and its kinds

The **plane of vibration** is the plane containing the direction of vibration of the electric vector and the direction of propagation. The **plane of polarisation** is the plane perpendicular to the plane of vibration, also containing the direction of propagation (it contains the magnetic vector).

3

Why light but not sound can be polarised.

KNOW FIRST

P1 Waves: the words you need T1 Polarisation and its kinds

Polarisation means selecting one direction of vibration perpendicular to the direction of travel, which is possible only for transverse waves. Light is transverse (E ⊥ direction of propagation), so it can be polarised. Sound in air is longitudinal (vibrations along the direction of travel), so there is no transverse direction to select.

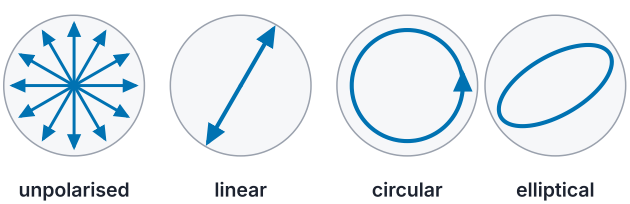
4

Unpolarised light and its representation.

KNOW FIRST

T1 Polarisation and its kinds

In unpolarised light the electric vector vibrates in all directions perpendicular to the direction of propagation, changing randomly and very rapidly (about 10^8 times a second), with no preferred direction. It is represented by a star of arrows in the transverse plane, or, in side view, by equal numbers of short lines (vibrations in the page) and dots (vibrations perpendicular to the page).



tip of the electric field E, seen with the light coming towards you

States of polarisation, drawn as the path traced by the tip of the electric field vector. Unpolarised light changes direction randomly and very fast; polarised light keeps an orderly pattern.

5

State Malus' law.

KNOW FIRST

T2 Polariser, analyser and Malus' law

When completely plane polarised light falls on an analyser, the intensity of the transmitted light is proportional to the square of the cosine of the angle θ between the plane of vibration of the incident light and the pass axis of the analyser: $I = I_0 \cos^2 \theta$.

6

Two polaroids at 60°.

KNOW FIRST

T2 Polariser, analyser and Malus' law

1. First polaroid: $I_1 = I_0/2$
   Unpolarised light is halved.
2. Second: $I = I_1 \cos^2 60^\circ = \frac{I_0}{2} \times \frac{1}{4} = \frac{I_0}{8}$
   Malus' law.

ANSWER $I = I_0/8$

7

Polaroid and its principle.

KNOW FIRST

T3 Polaroids, wire grids and scattering

A Polaroid is a thin sheet that produces plane polarised light. It contains long-chain molecules (polyvinyl alcohol with iodine) stretched so that they are aligned. It works by **dichroism** (selective absorption): the component of $\vec{E}$ parallel to the chains is absorbed because electrons can move along the chains; the perpendicular component is transmitted. The emergent light is plane polarised, along the pass axis (perpendicular to the chains).

8

Polarisation by scattering at 90°.

KNOW FIRST

T3 Polaroids, wire grids and scattering P2 Light is an electromagnetic wave

The electric field of the incident light makes the electrons of the molecules oscillate in the plane perpendicular to the beam. An oscillating charge radiates nothing along its line of oscillation. Seen at 90° to the beam, the oscillation component along the line of sight sends nothing to the observer; only the component perpendicular to both the beam and the line of sight is seen. So the scattered light is plane polarised.

9 PYQ N20 Q17

Brewster's law; Brewster window.

KNOW FIRST

T4 Polarisation by reflection: Brewster's law

Brewster's law: the tangent of the polarising angle equals the refractive index of the reflecting medium: $\tan i_B = n$. At this angle the reflected light is completely plane polarised (vibrations perpendicular to the plane of incidence) and the reflected and refracted rays are at right angles.

Brewster window: a glass plate set at the Brewster angle at the ends of a gas laser tube. Light vibrating in the plane of incidence passes through with no reflection loss, so the laser output is plane polarised and losses are minimised.

10

Polarising angle for glass, n = 1.5.

KNOW FIRST

T4 Polarisation by reflection: Brewster's law

1. $i_B = \tan^{-1}(1.5) = 56.3^\circ$

2. $r = 90^\circ - i_B = 33.7^\circ$
   Reflected ⊥ refracted. Check: $\sin 56.3^\circ / \sin 33.7^\circ = 0.832/0.555 = 1.50$ ✓

ANSWER $i_B = 56.3^\circ$, $r = 33.7^\circ$

11 PYQ N19 Q8 · adapted

Are both O and E rays plane polarised?

KNOW FIRST

T5 Double refraction: O-ray and E-ray

True. In double refraction both the ordinary and extraordinary rays are plane polarised, and their planes of vibration are perpendicular to each other: the E-ray vibrates in the principal section and the O-ray perpendicular to it. (Check: viewed through an analyser, the two images become dark alternately every 90°.)

12

Optic axis; uniaxial and biaxial.

KNOW FIRST

T5 Double refraction: O-ray and E-ray

The optic axis is a **direction** in a doubly refracting crystal along which the ordinary and extraordinary rays travel with the same velocity, so a ray along it is not doubly refracted. Crystals with one such direction (calcite, quartz, tourmaline) are **uniaxial**; those with two (mica, topaz) are **biaxial**.

13 PYQ N22 Q9 · adapted

Negative crystal: is v_o < v_e?

KNOW FIRST

T5 Double refraction: O-ray and E-ray T6 Huygens' explanation of double refraction

True. In a negative crystal (calcite) $\mu_e < \mu_o$, so the E-ray travels faster than the O-ray in all directions except along the optic axis, where both are equal. The E-wavelet (ellipsoid) lies outside the O-wavelet (sphere), touching it on the optic axis.

14

Principal section.

KNOW FIRST

T5 Double refraction: O-ray and E-ray

A principal section of a crystal is a plane containing the optic axis and perpendicular to a pair of opposite faces. When the plane of incidence is a principal section, the E-ray vibrates **in** the principal section and the O-ray vibrates **perpendicular** to it.

15 PYQ N24 Q9

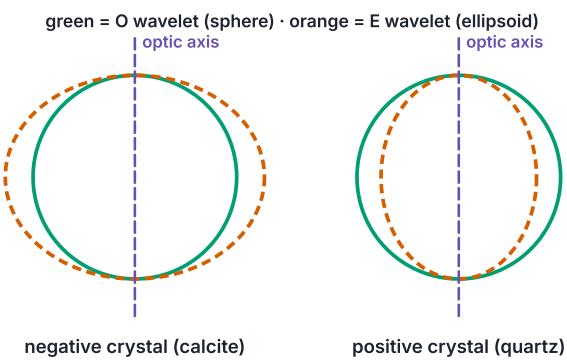
Huygens' explanation of double refraction.

KNOW FIRST

T6 Huygens' explanation of double refraction P3

Refraction, total internal reflection, Huygens' principle

According to Huygens, every point of a wavefront entering a doubly refracting crystal gives rise to two secondary wavelets: a **spherical** wavelet for the ordinary wave (same speed in all directions) and an **ellipsoidal** wavelet (ellipsoid of revolution about the optic axis) for the extraordinary wave. The two touch along the optic axis. The common tangent surfaces to the spheres and to the ellipsoids give the O and E wavefronts, so two refracted rays are produced.



Huygens' explanation. Each point of the wavefront sends out two wavelets: a sphere (O) and an ellipsoid of revolution (E). They touch along the optic axis, where both rays have the same speed. Negative crystal: E is faster off-axis ($\mu_e < \mu_o$); positive crystal: E is slower.

16

Optic axis perpendicular to the faces, normal incidence.

KNOW FIRST

T6 Huygens' explanation of double refraction

Only one image is seen: there is no double refraction. The light travels along the optic axis, where the spherical and ellipsoidal wavelets touch, so the O and E wavefronts coincide and both waves travel with the same speed c/μ_o in the same direction.

17

Elimination of the O-ray in a Nicol prism.

KNOW FIRST

T7 The Nicol prism P3 Refraction, total internal reflection, Huygens' principle

The O-ray travels from calcite ($\mu_o = 1.658$) to Canada balsam ($\mu = 1.55$), i.e. from denser to rarer. Its angle of incidence at the balsam layer is greater than the critical angle, so it is totally internally reflected to the blackened side and absorbed. The E-ray ($\mu_e = 1.486 < 1.55$) cannot undergo total internal reflection and is transmitted.

1. $\sin C = \dfrac{1.55}{1.658} = 0.935$, so $C = 69.2^\circ$

ANSWER C ≈ 69°

18 PYQ N20 Q16

What is a half-wave plate?

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave

A half-wave plate is a plate of doubly refracting crystal, cut with its optic axis parallel to its faces, of such thickness that it introduces a path difference of λ/2 (phase difference π) between the O and E waves: $d = \dfrac{\lambda}{2|\mu_o - \mu_e|}$. It turns plane polarised light at angle φ to its axis into plane polarised light at −φ (a rotation of 2φ), and reverses the sense of circularly polarised light.

19

Quarter-wave plate and its thickness.

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave

A quarter-wave plate is a doubly refracting plate, optic axis parallel to its faces, that introduces a path difference of λ/4 (phase difference π/2) between the O and E waves. From $(\mu_o - \mu_e)d = \lambda/4$:

$$d = \frac{\lambda}{4|\mu_o - \mu_e|}$$

It converts plane polarised light into circularly or elliptically polarised light, and vice versa.

20

Plane polarised at 45° on a QWP and on a HWP.

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave T8 Two perpendicular vibrations

The incident vibration splits into equal O and E components ($E_0 \cos 45^\circ = E_0 \sin 45^\circ$). **QWP:** they emerge with phase difference π/2 and equal amplitudes, so the light is **circularly polarised**. **HWP:** phase difference π, so the light is **plane polarised**, with the plane rotated through 2 × 45° = 90°.

21

Nicol prism vs Polaroid.

KNOW FIRST

T7 The Nicol prism T3 Polaroids, wire grids and scattering

Principle	Nicol: double refraction + total internal reflection. Polaroid: selective absorption (dichroism).
Quality	Nicol: almost perfect polarisation. Polaroid: slightly imperfect, a little coloured.
Size, cost	Nicol: small aperture, costly (calcite). Polaroid: large sheets, cheap.
Field	Nicol: narrow (about 28°). Polaroid: wide.

22

Right vs left circular light.

KNOW FIRST

T8 Two perpendicular vibrations

In circularly polarised light the tip of $\vec{E}$ moves on a circle. If, looking towards the oncoming light, it rotates **clockwise**, the light is **right** circularly polarised; **anticlockwise**, **left** circularly polarised. With $E_x = a\cos\omega t$, $E_y = a\cos(\omega t - \delta)$: δ = π/2 gives right and δ = 3π/2 (−π/2) gives left (Ghatak's convention). A half-wave plate changes one into the other.

23 PYQ N25 Q7

Optical activity.

KNOW FIRST

T12 Optical activity and Laurent's half-shade polarimeter

Optical activity is the rotation of the plane of vibration of plane polarised light as it passes through certain substances (quartz along its optic axis, sugar solution, turpentine). Substances rotating it clockwise (looking towards the source) are dextro-rotatory; anticlockwise, laevo-rotatory. The rotation is proportional to the thickness (path length) and, for solutions, to the concentration; it depends on wavelength and temperature. Fresnel explained it by the different speeds of right and left circular components.

24 PYQ N20 Q5

Define specific rotation.

KNOW FIRST

T12 Optical activity and Laurent's half-shade polarimeter

Specific rotation of an optically active substance, at a given temperature and wavelength, is the rotation produced by a 1 decimetre length of its solution of concentration 1 g/cm³:

$$[\alpha]_{\lambda}^{T} = \frac{\theta}{l\,c} \quad (l \text{ in dm},\ c \text{ in g/cm}^3)$$

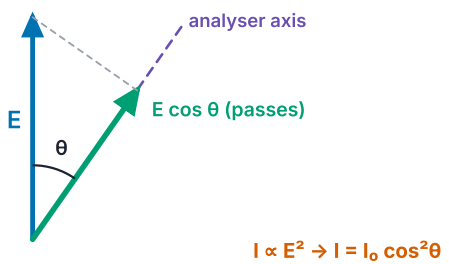
Section B · Paragraph / problem
5 marks each.

25

Derive Malus' law; three polaroids.

KNOW FIRST

T2 Polariser, analyser and Malus' law M1 Splitting an arrow into two parts P1 Waves: the words you need



Why $\cos^2$: split E into a part along the analyser's axis ($E\cos\theta$, which passes) and a part across it (blocked). Intensity goes as amplitude squared.

1. Plane polarised light of amplitude E_0 falls on an analyser whose pass axis makes angle θ with the vibration
2. Resolve: $E_0\cos\theta$ along the pass axis (transmitted), $E_0\sin\theta$ perpendicular (blocked)
3. Intensity ∝ amplitude²: $I = k(E_0\cos\theta)^2 = I_0\cos^2\theta$, with $I_0 = kE_0^2$

Three polaroids:

1. After P_1: $I_0/2$ (unpolarised halved)
2. After P_2 (45°): $\frac{I_0}{2}\cos^2 45^\circ = \frac{I_0}{4}$
3. After P_3 (45° to P_2): $\frac{I_0}{4}\cos^2 45^\circ = \frac{I_0}{8}$

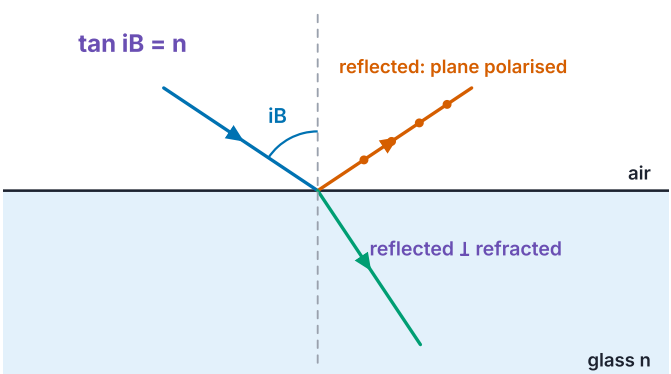
ANSWER $I = I_0\cos^2\theta$; **three polaroids give** $I_0/8$ **(zero without the middle one)**

26

Brewster's law; reflected ⊥ refracted.

KNOW FIRST

T4 Polarisation by reflection: Brewster's law P3 Refraction, total internal reflection, Huygens' principle M2 Trig identities you will use



Brewster's law. At the angle i_B with $\tan i_B = n$, the reflected light is completely plane polarised (vibrations perpendicular to the plane of incidence: the dots), and the reflected and refracted rays are at 90°.

At the polarising angle the reflection coefficient for the component vibrating in the plane of incidence vanishes (Ghatak §24.5):

1. $r_{\parallel} = \dfrac{n_2 \cos i - n_1 \cos r}{n_2 \cos i + n_1 \cos r} = 0 \Rightarrow n_2 \cos i = n_1 \cos r$
   Fresnel's formula for the parallel component.
2. Snell: $n_1 \sin i = n_2 \sin r$
3. Multiply the two: $n_1 n_2 \sin i \cos i = n_1 n_2 \sin r \cos r \Rightarrow \sin 2i = \sin 2r$
4. Since $i \neq r$: $2i = 180^\circ - 2r$, so $i + r = 90^\circ$
   Reflected and refracted rays perpendicular: angle between them $= 180^\circ - i - r = 90^\circ$.
5. Then $\sin r = \cos i$, and Snell gives $n_1 \sin i_B = n_2 \cos i_B$, so $\tan i_B = \dfrac{n_2}{n_1} = n$

Physical picture: the refracted wave's electrons oscillate (for the parallel component) along the direction of the would-be reflected ray, and an oscillating dipole does not radiate along its axis, so the reflected light lacks that component and is plane polarised perpendicular to the plane of incidence.

ANSWER $\tan i_B = n$ **and** $i_B + r = 90^\circ$

27

Pile of plates.

KNOW FIRST

T4 Polarisation by reflection: Brewster's law

A pile of plates is 10–15 thin glass plates stacked parallel, with unpolarised light incident at the polarising angle ($\tan i_B = n$).

1. At every surface, the reflected light is plane polarised perpendicular to the plane of incidence (Brewster)
2. Only about 15% of that perpendicular component is reflected per surface ($n = 1.5$); the parallel component is fully transmitted
3. After many surfaces almost all the perpendicular vibration has been reflected away, so the **transmitted** beam is nearly plane polarised, vibrating in the plane of incidence

Why the transmitted light is used: the reflected light from one surface is polarised but very weak (about 15% of one component); the transmitted beam is much more intense and, after many plates, nearly completely polarised.

28 PYQ N20 Q28

Quartz HWP and QWP, λ = 5000 Å, μE = 1.553, μO = 1.544.

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave M4

Small numbers: Å, nm, μm

1. $|\mu_e - \mu_o| = 0.009$; λ = $5000 \times 10^{-10} = 5 \times 10^{-7}$ m
2. HWP: $d = \dfrac{\lambda}{2|\mu_e - \mu_o|} = \dfrac{5 \times 10^{-7}}{0.018} = 2.78 \times 10^{-5}$ m

3 QWP: $d = \dfrac{\lambda}{4|\mu_e - \mu_o|} = \dfrac{5 \times 10^{-7}}{0.036} = 1.39 \times 10^{-5}$ m

ANSWER **HWP 27.8 µm (0.0278 mm); QWP 13.9 µm (0.0139 mm)**

SAME IDEA, OTHER YEARS

N19 Q30 / N22 Q30 (λ = 5500 Å, µE = 1.54, µO = 1.53): $d = \dfrac{5.5 \times 10^{-7}}{4 \times 0.01} = 1.375 \times 10^{-5}$ m = **13.75 µm.**

N25 Q17 (589 nm, 1.544 and 1.553): $d = \dfrac{5.89 \times 10^{-7}}{4 \times 0.009} = 1.64 \times 10^{-5}$ m = **16.4 µm.**

29 PYQ N25 Q16

E-ray velocity at 45° to the optic axis.

KNOW FIRST

T5 Double refraction: O-ray and E-ray T6 Huygens' explanation of double refraction

1 The E-wavelet is an ellipsoid; in polar form $\dfrac{1}{v_e^2} = \dfrac{\cos^2\theta}{v_\parallel^2} + \dfrac{\sin^2\theta}{v_\perp^2}$

$v_\parallel = 1.81 \times 10^8$ along the axis, $v_\perp = 2.02 \times 10^8$ perpendicular.

2 θ = 45°: $\dfrac{1}{v_e^2} = \dfrac{1}{2}\left[\dfrac{1}{(1.81)^2} + \dfrac{1}{(2.02)^2}\right] \times 10^{-16} = \dfrac{1}{2}(0.3052 + 0.2451) \times 10^{-16}$

3. $= 0.2752 \times 10^{-16}$ s²/m², so $v_e = \dfrac{1}{\sqrt{0.2752}} \times 10^8 = 1.906 \times 10^8$ m/s

ANSWER $v_e \approx 1.91 \times 10^8$ **m/s (between the two extremes)**

30

E-ray obeys Snell's law when the optic axis ⊥ plane of incidence.

KNOW FIRST

T6 Huygens' explanation of double refraction P3

Refraction, total internal reflection, Huygens' principle

1. Plane wavefront AB meets the crystal surface obliquely at angle i. While the wave travels BC $= ct$ in air, A sends wavelets into the crystal
2. The optic axis is perpendicular to the plane of incidence (the page). The E-ellipsoid is a figure of revolution about that axis, so its section by the page is a **circle**, of radius $v_\perp t = (c/\mu_e)\, t$
3. The E-wavefront is the tangent from C to this circle, exactly as for an ordinary isotropic medium of speed c/μ_e
4. From the triangles: $\sin i = \dfrac{ct}{AC}$, $\sin r_e = \dfrac{(c/\mu_e)t}{AC}$, so $\dfrac{\sin i}{\sin r_e} = \mu_e$

   The E-ray behaves like an ordinary ray of index μ_e; the O-ray obeys $\sin i / \sin r_o = \mu_o$.

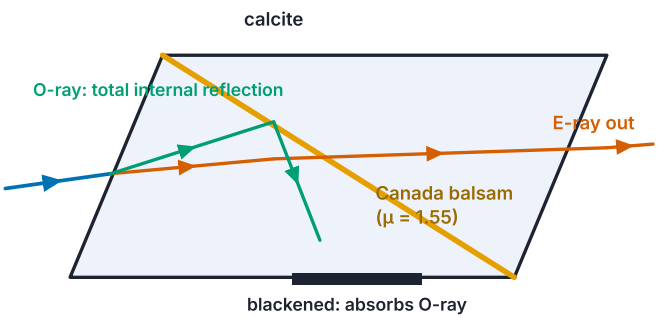
This is how μ_e is measured: with a prism cut so that the optic axis is perpendicular to the plane of incidence.

31

Construction and working of a Nicol prism.

KNOW FIRST

T7 The Nicol prism T5 Double refraction: O-ray and E-ray P3 Refraction, total internal reflection, Huygens' principle



Nicol prism. A calcite crystal is cut along a diagonal and cemented with Canada balsam. The O-ray ($\mu_o = 1.658$) meets the balsam (1.55) beyond the critical angle and is totally reflected to the blackened side. The E-ray ($\mu_e = 1.486 < 1.55$) cannot be totally reflected and passes out, plane polarised.

Construction: a calcite crystal whose length is about three times its breadth. Its end faces are ground so that their angle with the long edges changes from 71° to 68°. It is cut into two along a plane perpendicular to the principal section and to the new end faces, and the two halves are cemented with Canada balsam ($\mu = 1.55$). The sides are blackened.

Working:

1. Unpolarised light entering parallel to the long edge splits into the O-ray ($\mu_o = 1.658$) and the E-ray ($\mu_e = 1.486$)
2. For the O-ray, balsam is rarer (1.55 < 1.658). Critical angle $C = \sin^{-1}(1.55/1.658) \approx 69°$. Its angle of incidence exceeds C, so it is totally reflected and absorbed at the blackened side
3. For the E-ray, balsam is denser (1.55 > 1.486), so no total reflection; it emerges plane polarised, vibrating in the principal section

Use: as a polariser and an analyser. Between two Nicols, rotating the second gives maximum light when parallel and extinction when crossed.

32

Plate between crossed polarisers.

KNOW FIRST

T10 Producing plane, circular and elliptical light T9
Wave plates: quarter-wave and half-wave M1
Splitting an arrow into two parts

1. After P_1: amplitude A, intensity $I_1 = I_0/2$. The plate's axis is at φ: components $A\cos\phi$ (E) and $A\sin\phi$ (O)
2. After the plate the O component has an extra phase δ. The analyser P_2 (⊥ P_1) takes $A\cos\phi\sin\phi$ from one and $-A\sin\phi\cos\phi\, e^{i\delta}$ from the other

3. Resultant amplitude $A \sin\phi \cos\phi\,(1 - e^{i\delta})$; $|1 - e^{i\delta}|^2 = 2 - 2\cos\delta = 4\sin^2(\delta/2)$

4. $I = I_1 \cdot 4\sin^2\phi\cos^2\phi\sin^2\frac{\delta}{2} = I_1 \sin^2 2\phi\, \sin^2\frac{\delta}{2}$

5. φ = 45°: QWP (δ = π/2): $I = I_1/2 = I_0/4$. HWP (δ = π): $I = I_1 = I_0/2$

ANSWER QWP: $I_0/4$; HWP: $I_0/2$

33

Left circular light through a calcite plate of 0.005141 mm.

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave T8 Two perpendicular vibrations M4 Small numbers: Å, nm, μm

1. $$\delta = \frac{2\pi}{\lambda}(\mu_o - \mu_e)d = \frac{2\pi \times 0.17195 \times 5.141 \times 10^{-6}}{5.893 \times 10^{-7}}$$

2. $= 2\pi \times 1.500 = 3\pi$
   An odd multiple of π: the plate acts as a half-wave plate.

3. A half-wave plate adds a phase π between the components, which reverses the sense of rotation

ANSWER δ = 3π; the light emerges right circularly polarised

34

Plane = right + left circular; Fresnel's theory; rotation in 1 mm quartz.

KNOW FIRST

T12 Optical activity and Laurent's half-shade polarimeter T8 Two perpendicular vibrations M2 Trig identities you will use

1. Right circular: $E_x = E_0 \cos \omega t$, $E_y = E_0 \sin \omega t$. Left circular: $E_x = E_0 \cos \omega t$, $E_y = -E_0 \sin \omega t$

2. Sum: $E_x = 2E_0 \cos \omega t$, $E_y = 0$: plane polarised along x

   The y parts cancel at every instant.

3. In an active medium the two circular waves travel with indices $n_R \neq n_L$. After length z, their phases differ: one is ahead by $\frac{2\pi}{\lambda}(n_L - n_R)z$

4. Adding them again gives plane polarised light whose direction has turned by half that phase difference: $\phi = \frac{\pi}{\lambda}(n_L - n_R)z$

5. Quartz: $\phi = \frac{\pi \times 7 \times 10^{-5} \times 10^{-3}}{6 \times 10^{-7}} = 0.367$ rad $= 21^\circ$

ANSWER Rotation ≈ 21° per mm of quartz

35 PYQ N20 Q23

Producing plane, circular and elliptical light.

KNOW FIRST

T10 Producing plane, circular and elliptical light T9 Wave plates: quarter-wave and half-wave T7 The Nicol prism T8 Two perpendicular vibrations

Plane polarised light: pass monochromatic light through a Nicol prism (or Polaroid). The emergent light vibrates in the principal section of the Nicol.

Circularly polarised light: let this plane polarised light fall normally on a quarter-wave plate with its optic axis at 45° to the vibration. Components $E_0 \cos 45^\circ$ (E) and $E_0 \sin 45^\circ$ (O) are equal; the plate introduces δ = π/2. With $E_x = a\cos\omega t$, $E_y = a\cos(\omega t - \pi/2) = a\sin\omega t$: $E_x^2 + E_y^2 = a^2$, a circle.

Elliptically polarised light: as above, but with the optic axis at an angle φ ≠ 0°, 45°, 90°. Amplitudes $a = E_0\cos\phi$, $b = E_0\sin\phi$ are unequal; δ = π/2 gives $\frac{E_x^2}{a^2} + \frac{E_y^2}{b^2} = 1$, an ellipse with axes along the plate's axes. (Any plate with δ ≠ 0, π/2, π also gives elliptical light.)

Marks: plane 1 · circular with reason 2 · elliptical with reason 2.

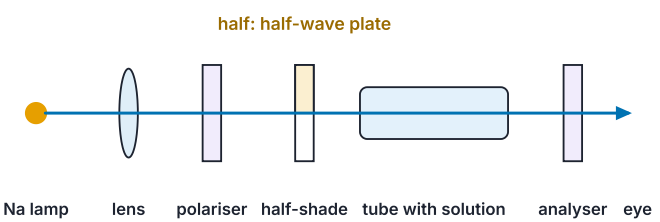
36 PYQ N24 Q18

Double refraction; Laurent's half-shade polarimeter.

KNOW FIRST

T5 Double refraction: O-ray and E-ray T12 Optical activity and Laurent's half-shade polarimeter T9 Wave plates: quarter-wave and half-wave

Double refraction: when unpolarised light enters an anisotropic crystal such as calcite, it splits into two refracted rays, the ordinary ray (obeys Snell's law, same speed in all directions) and the extraordinary ray (speed depends on direction). Both are plane polarised at right angles to each other.



Laurent's half-shade polarimeter. One half of the half-shade disc is a half-wave plate, the other half plain glass. With the analyser set for equal dimness of both halves, any rotation by the solution upsets the balance; turning the analyser back to equal dimness measures the rotation.

Laurent's half-shade polarimeter measures the rotation of the plane of polarisation by a solution.

1. **Parts:** sodium lamp, collimating lens, polarising Nicol, half-shade plate, tube for the solution, analysing Nicol with telescope and circular scale

2. **Half-shade plate:** one semicircle is a quartz half-wave plate (optic axis parallel to the dividing line); the other is glass absorbing equally. If the polariser's vibration OA makes a small angle ε with the axis, light through the HWP vibrates along OB, at −ε

3. When the analyser is perpendicular to the bisector of AOB, both halves are **equally dim**. A slight turn makes one half brighter and the other darker, so this setting is very sensitive

4. **Procedure:** with the tube full of water, set equal dimness; read the scale. Fill with solution: the planes rotate by θ and the balance is lost. Turn the analyser until equal dimness returns. The difference of readings is θ

5. Specific rotation $[\alpha] = \dfrac{\theta}{l\,c}$ (l in dm, c in g/cm³); or concentration $c = \dfrac{\theta}{[\alpha] l}$

37

Calcite QWP for 4046 Å used at 7065 Å.

KNOW FIRST

T9 Wave plates: quarter-wave and half-wave T8 Two perpendicular vibrations M4 Small numbers: Å, nm, µm

1. $d = \dfrac{4.046 \times 10^{-7}}{4 \times 0.18440} = 5.49 \times 10^{-7}$ m $= 0.549$ µm
2. At 7065 Å: $\delta = \dfrac{2\pi \times 0.16848 \times 5.485 \times 10^{-7}}{7.065 \times 10^{-7}} = 2\pi \times 0.1308 = 0.82$ rad ≈ 47°
3. δ ≠ π/2, so equal components at 45° do **not** give a circle: the light is **elliptically** polarised (axes at 45° to the plate's axes)

ANSWER **d ≈ 0.55 µm; at 7065 Å δ ≈ 47°, so elliptical light emerges**

38

Partially polarised light; I_max = 3 I_min.

KNOW FIRST

T2 Polariser, analyser and Malus' law T1 Polarisation and its kinds

Partially polarised light is a mixture of unpolarised and plane polarised light. Through a rotating analyser its intensity varies between I_{max} and I_{min} without ever becoming zero.

1. $P = \dfrac{I_{max} - I_{min}}{I_{max} + I_{min}} = \dfrac{3-1}{3+1} = 0.5$

2. Unpolarised part I_u gives $I_u/2$ at every angle; polarised part I_p adds $I_p \cos^2\theta$. So $I_{min} = I_u/2$, $I_{max} = I_u/2 + I_p$

3. $I_{max} = 3I_{min}$ gives $I_p = 2I_{min} = I_u$
Equal amounts of polarised and unpolarised light.

ANSWER **Degree of polarisation 50%: half polarised, half unpolarised**

Section C · Essays
10 marks each. Theory 3–4 · derivation/figure 3–4 · method/result 2–3.

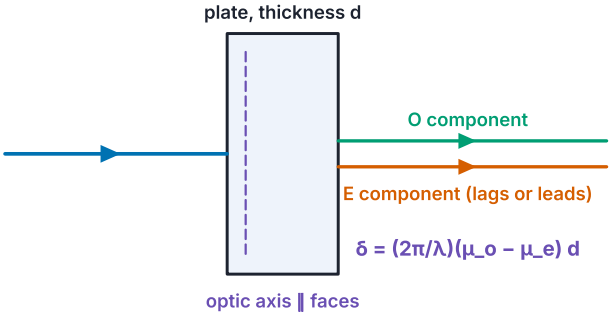
39 PYQ N19 Q34

Produce circularly and elliptically polarised light; analyse the type of polarisation.

KNOW FIRST
T10 Producing plane, circular and elliptical light T11 Analysing polarised light T9 Wave plates: quarter-wave and half-wave T8 Two perpendicular vibrations T7 The Nicol prism

1. Theory: two perpendicular vibrations

If plane polarised light of amplitude E_0 falls normally on a crystal plate (optic axis parallel to the faces) with its vibration at angle φ to the optic axis, it splits into E and O vibrations $a = E_0 \cos\phi$ and $b = E_0 \sin\phi$ that travel at different speeds. They emerge with phase difference $\delta = \frac{2\pi}{\lambda}(\mu_o - \mu_e)d$ and combine to $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta.$



A wave plate. Light enters at 90° to the optic axis, so the O and E vibrations travel along the same line but at different speeds. Leaving the plate, one is behind the other by a phase δ. Quarter-wave plate: δ = π/2. Half-wave plate: δ = π.

2. Production of circularly polarised light

Monochromatic light is made plane polarised by a Nicol N_1 and falls on a quarter-wave plate ($d = \lambda/4|\mu_o - \mu_e|$, δ = π/2) with its optic axis at 45° to the vibration. Then $a = b = E_0/\sqrt{2}$ and δ = π/2 give $x^2 + y^2 = a^2$: circularly polarised light. (Check: rotating a Nicol behind it shows no change in intensity.)

3. Production of elliptically polarised light

Same arrangement with the optic axis at an angle φ other than 0°, 45°, 90°: $a \neq b$, δ = π/2, giving $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, an ellipse with axes along the optic axis and perpendicular to it.

4. Analysis of the given light

Examine the beam through a rotating Nicol (analyser):

1. **Two positions of zero intensity** per rotation → plane polarised

2. **No variation** → unpolarised, circular, or a mixture. Insert a QWP before the analyser and rotate the analyser:
   (a) still no variation → unpolarised; (b) two zeros → circularly polarised (QWP adds π/2 to the existing π/2, giving plane light); (c) variation without zero → mixture of unpolarised and circular.

3. **Variation, but no zero** → elliptical, or a mixture of plane and unpolarised. Place the QWP with its optic axis parallel to the analyser's position of maximum intensity, then rotate the analyser:
   (a) two zeros → elliptically polarised (axes of ellipse along plate axes, δ becomes 0 or π); (b) no zero, maximum at the same position → plane + unpolarised; (c) no zero, maximum shifted → elliptical + unpolarised.

5. Result table

ANALYSER ALONE	WITH QWP	LIGHT IS
two zeros	not needed	plane polarised
constant	constant	unpolarised
constant	two zeros	circular
varies, no zero	two zeros	elliptical
varies, no zero	no zero, same max	plane + unpolarised

Covers also the last part of N20 Q34 ("How would you produce and detect circularly polarised light?").

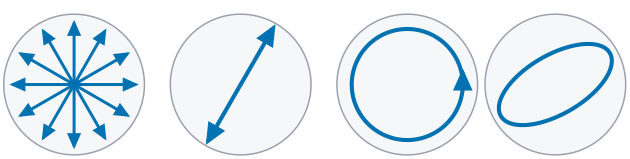
40 PYQ N20 Q34 (first two parts)

Distinguish plane, circular, elliptical polarisation; production of linearly polarised light by reflection and refraction.

KNOW FIRST

T1 Polarisation and its kinds T4 Polarisation by reflection: Brewster's law T5 Double refraction: O-ray and E-ray T7 The Nicol prism T8 Two perpendicular vibrations

1. The three kinds



tip of the electric field E, seen with the light coming towards you

States of polarisation, drawn as the path traced by the tip of the electric field vector. Unpolarised light changes direction randomly and very fast; polarised light keeps an orderly pattern.

Plane E vibrates along one fixed line perpendicular to the direction of propagation. Two perpendicular components in phase ($\delta = 0$ or π).

Circular E has constant magnitude and its tip rotates uniformly on a circle. Two equal perpendicular components with $\delta = \pi/2$.

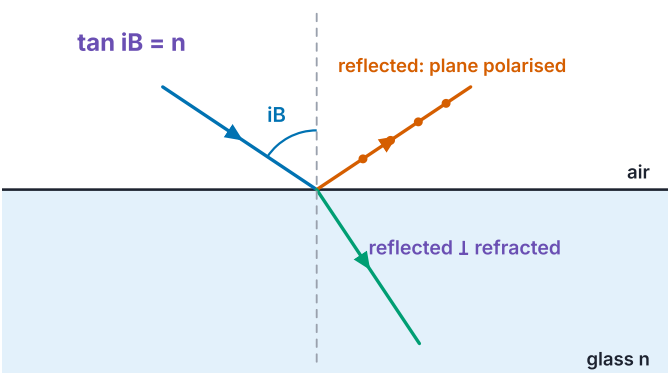
Elliptical The tip of E traces an ellipse; magnitude and direction both vary. Unequal

components with δ = π/2, or any components with δ ≠ 0, π/2, π.

Tests: through a rotating analyser, plane light gives two extinctions; circular gives constant intensity; elliptical gives a variation without extinction.

2. By reflection (Brewster)

Unpolarised light reflected from glass at the polarising angle i_B, where $\tan i_B = n$, is completely plane polarised with vibrations perpendicular to the plane of incidence. At this angle the reflected and refracted rays are perpendicular. For glass ($n = 1.5$), $i_B = 56.3°$.



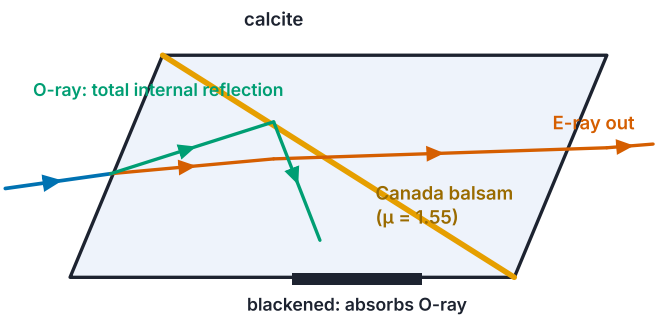
Brewster's law. At the angle i_B with $\tan i_B = n$, the reflected light is completely plane polarised (vibrations perpendicular to the plane of incidence: the dots), and the reflected and refracted rays are at 90°.

3. By refraction

(a) Pile of plates: the light refracted through one plate at i_B is only partly polarised (it has lost about 15% of its perpendicular component).

Through 10–15 plates at i_B, the perpendicular component is reflected away at every surface; the transmitted light becomes almost completely plane polarised in the plane of incidence.

(b) Double refraction: in calcite, unpolarised light splits into O and E rays, plane polarised at right angles. Removing one gives plane polarised light. In the **Nicol prism**, the O-ray is totally reflected at the Canada balsam layer (C ≈ 69°) and absorbed, and the E-ray emerges plane polarised.



Nicol prism. A calcite crystal is cut along a diagonal and cemented with Canada balsam. The O-ray ($\mu_o = 1.658$) meets the balsam (1.55) beyond the critical angle and is totally reflected to the blackened side. The E-ray ($\mu_e = 1.486 < 1.55$) cannot be totally reflected and passes out, plane polarised.

(c) Dichroism: crystals like tourmaline, or Polaroid sheets, absorb one of the two vibrations strongly and transmit the other.

Marks: distinctions with figure 3 · reflection with Brewster's law 3 · refraction (pile of plates, Nicol) 4.

41

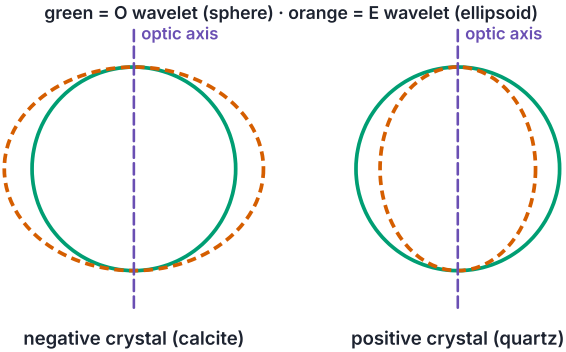
Huygens' theory of double refraction; wave surfaces; three cases of normal incidence.

KNOW FIRST

T6 Huygens' explanation of double refraction T5 Double refraction: O-ray and E-ray P3 Refraction, total internal reflection, Huygens' principle

1. Postulates

1. A wavefront entering a uniaxial crystal produces two wavefronts, ordinary and extraordinary.
2. For the O-wave the speed is the same in all directions: the secondary wavelet is a **sphere**.
3. For the E-wave the speed varies with direction: the wavelet is an **ellipsoid of revolution** about the optic axis.
4. Along the optic axis both speeds are equal (c/μ_o): sphere and ellipsoid touch there.
5. Negative crystal (calcite, $\mu_e < \mu_o$): ellipsoid outside the sphere. Positive crystal (quartz, $\mu_e > \mu_o$): inside.

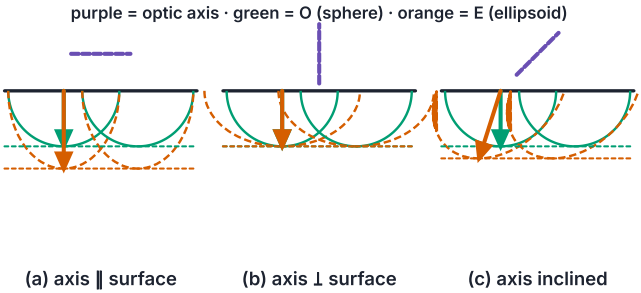


Huygens' explanation. Each point of the wavefront sends out two wavelets: a sphere (O) and an ellipsoid of revolution (E). They touch along the optic axis, where both rays have the same speed. Negative crystal: E is faster off-axis ($\mu_e < \mu_o$); positive crystal: E is slower.

2. Construction

From points on the surface reached by the incident wavefront, draw the sphere and ellipsoid reached in time t. The plane touching all the spheres is the O-wavefront; the plane touching all the ellipsoids is the E-wavefront. Each ray joins the wavelet's centre to its point of contact.

3. Normal incidence, three orientations



Huygens' construction for normal incidence on calcite. (a) Both wavefronts parallel, rays along the same line, different speeds. (b) Wavefronts coincide: no double refraction. (c) The E-ray goes to the point where its wavefront touches the ellipsoid, so it is bent sideways.

1. **(a) Optic axis parallel to the surface** (in the plane of incidence): sphere radius $(c/\mu_o)t$, ellipsoid semi-axis $(c/\mu_e)t$ normal to the surface. Both tangent planes are parallel to the surface, and both rays go straight down along the same line, but the wavefronts are separated: the waves travel with different speeds. No separation of rays, but a phase difference: the principle of wave plates

2. **(b) Optic axis perpendicular to the surface:** the sphere and ellipsoid touch along the normal, so the two wavefronts coincide: no double refraction; one wave with speed c/μ_o

3. **(c) Optic axis inclined to the surface:** the O-ray goes straight along the normal. The E-wavefront is still parallel to the surface, but it touches each ellipsoid at a point that is not straight below its centre, so the E-ray is bent sideways: two separate rays and two images, even at normal incidence

4. Conclusion

Huygens' theory explains the existence of two rays, the absence of double refraction along the optic axis, the deviation of the E-ray at normal incidence, and (with the axis perpendicular to the plane of incidence) the fact that the E-ray then obeys Snell's law with index μ_e. The perpendicular polarisation of the two rays came later from the electromagnetic theory.

Marks: postulates 3 · wave surfaces figure 2 · three cases with figures 4 · conclusion 1.

42

Resultant of two perpendicular SHMs with phase difference δ.

KNOW FIRST

T8 Two perpendicular vibrations M2 Trig identities you will use M3 Two vibrations, a phase difference, and an ellipse

1. Setting up

Two vibrations of the same frequency along perpendicular directions (the E and O components

of light):

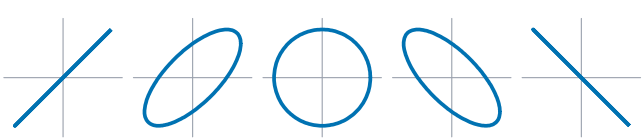
$$x = a\cos\omega t, \qquad y = b\cos(\omega t - \delta)$$

2. Eliminating time

1. $\cos\omega t = x/a$, $\sin\omega t = \sqrt{1 - x^2/a^2}$
2. $\frac{y}{b} = \cos\omega t\cos\delta + \sin\omega t\sin\delta = \frac{x}{a}\cos\delta + \sqrt{1 - \frac{x^2}{a^2}}\sin\delta$
3. $\left(\frac{y}{b} - \frac{x}{a}\cos\delta\right)^2 = \left(1 - \frac{x^2}{a^2}\right)\sin^2\delta$
4. $\frac{x^2}{a^2}\cos^2\delta + \frac{x^2}{a^2}\sin^2\delta + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta$

ANSWER $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{2xy}{ab}\cos\delta = \sin^2\delta$**: in general an ellipse**

3. Special cases



δ = 0 δ = π/4 δ = π/2 δ = 3π/4 δ = π

x = a cos ωt, y = a cos(ωt − δ): line → ellipse → circle → ellipse → line

Adding two perpendicular vibrations of equal amplitude with phase difference δ. δ = 0 or π: a straight line (linear). δ = π/2: a circle (circular). In between: ellipses. With unequal amplitudes, δ = π/2 gives an ellipse with axes along x and y.

1. **δ = 0:** $\left(\frac{x}{a} - \frac{y}{b}\right)^2 = 0$, i.e. $y = \frac{b}{a}x$: a straight line through the origin → **plane polarised**

2. **δ = π:** $y = -\frac{b}{a}x$: a straight line the other way → **plane polarised**

3. **δ = π/2:** $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$: an ellipse with axes along x and y → **elliptically polarised**; if $a = b$: $x^2 + y^2 = a^2$ → **circularly polarised**

4. **General δ:** an oblique ellipse inscribed in the rectangle $2a \times 2b$ → **elliptically polarised**. Its axes are tilted by ψ with $\tan 2\psi = \frac{2ab\cos\delta}{a^2 - b^2}$

4. Sense of rotation

At $t = 0$: $(x, y) = (a, b\cos\delta)$; a moment later y increases if $\sin\delta > 0$. So for $0 < \delta < \pi$ the tip turns one way (right-handed in Ghatak's convention, looking at the oncoming wave) and for $\pi < \delta < 2\pi$ the other way (left-handed).

5. Application

A crystal plate provides exactly such components with a controllable δ: a QWP gives δ = π/2 (circular or elliptical light), a HWP gives δ = π (plane light rotated by 2φ). So all three kinds of polarisation are produced from plane polarised light by wave plates.

Marks: derivation 4 · cases with figures 4 · application/sense 2.